

Jeffrey Watchorn Ph.D.

ENGAGEMENT MANAGER · AI FOR ENERGY · RESTLESS ADVOCATE FOR CUSTOMER SUCCESS

167 McRoberts Avenue, Toronto, Ontario, Canada

☎ (+1) 226-791-6704 | ✉ jeffreydwatc@outlook.com | 🌐 jwatchorn.github.io | 📧 jwatchorn | 📄 jeffreydwatc | 📧 Jeffrey Watchorn

Technical Skills

Program Management: End-to-End SaaS Delivery, Roadmap Execution, Dependency and Risk Management, Client Engagement
Communication: Technical Writing (Journals, Patents) and Presentation (Stakeholders, Public), French proficiency (B2)
Programming: Python (Pandas, Numpy, Scikit-learn, TensorFlow, Pytorch), SQL, Git, Jupyter, Jira

Relevant Work Experience

Validere Technologies

Toronto, Canada

ENGAGEMENT MANAGER

April 2025 - Present

- Owned end-to-end delivery of SAAS client engagements for a portfolio of global energy companies (North America and Oceania, 2 million CAD ARR), serving as primary point of contact and accountable owner for project roadmaps, milestones, and on-time delivery
- Consistently achieved between 5-10% ARR growth through new use case discovery including developing and delivering Validere's first set of Agentic features
- Coordinated cross-functional teams to deliver SaaS and AI functionality, using technical accumen support feature delivery, validation, and reporting while managing scope, risks, and dependencies

Acceleration Consortium

Toronto, Canada

STAFF RESEARCH SCIENTIST

Jan. 2024 - April 2025

- As the first research scientist, I designed, created, and managed an autonomous, AI-powered laboratory for drug formulations discovery.
- Developed laboratory data analysis and machine learning pipelines (Regression, Classification, RAG and LLM) to recommend experiments based on molecular features, remotely execute these experiments, and store experimental data.
- Secured and managed a large (budget: 1 million CAD) sponsored industrial research project for discovery of colloidal materials, routinely communicated experimental findings and project updates to key external stakeholders

Statistics Canada

Ottawa, Canada

DATA ANALYST (CONTRACT)

May. 2017 - Sept. 2017

- Developed ETL pipelines for accessing and analyzing Statistics Canada databases used for national indicators (such as consumer price index).
- Implemented machine learning tools, including KMeans clustering, for both exploratory data analysis and to dynamically assign product category labels to sample consumer shopping carts

Education

Doctor of Philosophy in Chemical Engineering

Toronto, Ontario, Canada

UNIVERSITY OF TORONTO, FRANK GU LAB

Sept. 2018 - Sept. 2023

- Machine learning enabled discovery for next-generation biomaterials.** Originated DISCO NMR to map materials-protein interactions with atomic precision. I used this data to develop a machine learning framework to power a self-driving materials discovery platform.

Bachelor of Applied Science in Nanotechnology Engineering

Waterloo, Ontario, Canada

UNIVERSITY OF WATERLOO, HONOURS, CO-OPERATIVE PROGRAM, WITH DISTINCTION

Sept. 2012 - May. 2017

Selected Publications

Nazeri, M.*; Mei, S.*; Watchorn, J.*; et al. *RAISE: a self-driving laboratory for interfacial property formulation discovery*. RSC Digital Discovery (2026). [full text at publisher](#)

Ansari, M.; Watchorn, J.; Brown, J. S.; Brown, C. E. *dZiner: Rational Inverse Design of Materials with AI Agents*. AI4Mat-NeurIPS-2024 (2024), Spotlight Article. [OpenReview|arXiv](#)

Stuart, S.*; Watchorn, J.*; Gu, F. X. *An Interpretable Machine Learning Framework for Modelling Macromolecular Interaction Mechanisms with Nuclear Magnetic Resonance*. RSC Digital Discovery (2023). Cover Article. [full text at publisher](#)

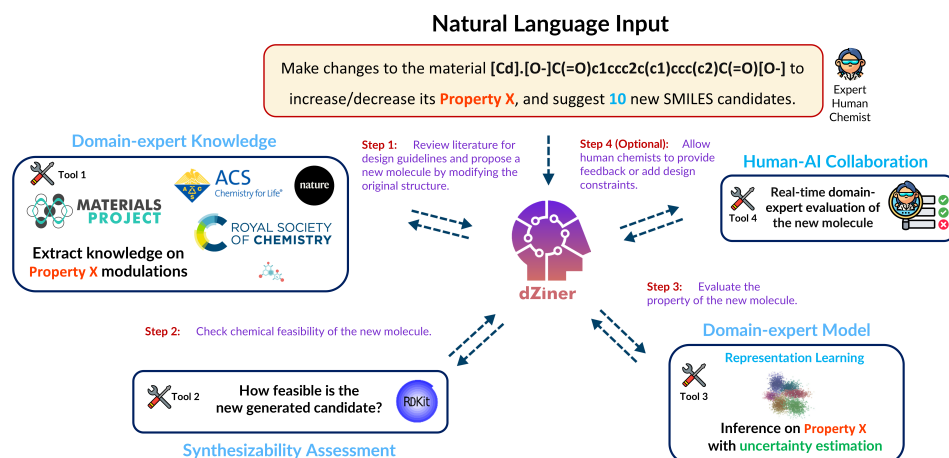
Stuart, S.; Watchorn, J.; Gu, F. X. *Sizing up feature descriptors for macromolecular machine learning with polymeric biomaterials*. npj Computational Materials, 2023, 9, 102. [full text at publisher](#)

Honors & Awards

2023	University of Toronto Chair's Discovery Award , Award - Research Excellence	Toronto, Canada
2019, 2021, 2022	Queen Elizabeth II Graduate Scholarship in Science and Technology , Scholarship - Provincial	Toronto, Canada
2020, 2021	Mclean Foundation Graduate Scholarships In Science And Technology , Scholarship - Institutional	Toronto, Canada

Portfolio

dZiner: Rational Inverse Design of Materials with AI Agents dZiner is an agentic AI that we built for NeurIPS AI-4MAT using **LangChain**. The AI agent uses the **REACT** framework to solve molecular design problems, by first interpreting a design task, leveraging **RAG** to review scientific literature and parse design guidelines, modifying the initial SMILES string, and finally evaluating its own performance. The evaluation is done by evaluating the candidate molecule using existing task-specific machine learning and physical models (including: **GCNNs**, **CGCNNs**, and **AutoDock Vina**). Along with Langchain, I worked on the surfactant design task conducting **model evaluation**, **benchmarking** and **prompt engineering**.

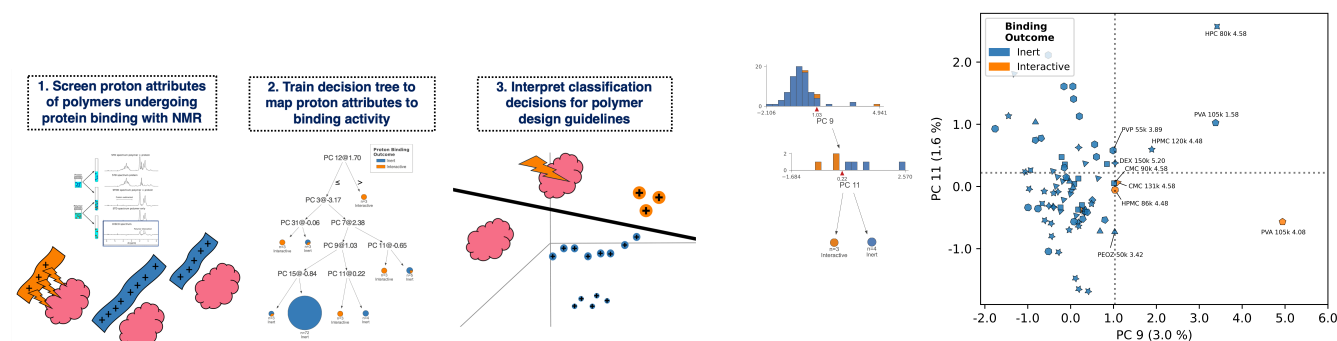


dZiner parses expert knowledge and uses chemistry tools to design molecules

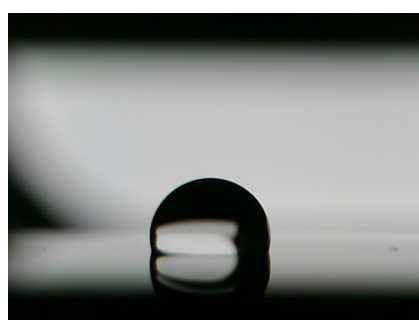


After presenting at NeurIPS!

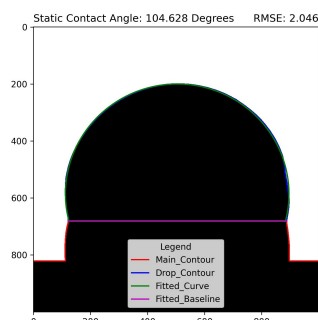
INFRNO: Interpretable framework for uncovering interaction opportunities in macromolecules Infrno leverages tree-based models (**RandomForest** and **DecisionTree**) to interpret high fidelity spectroscopy data with atomic precision. By training a binding **classifier** on a dataset that has been decorrelated with **PCA** I uncovered key insights into the structure-property relationships that underpin polymer-mucin binding. By analyzing the **factor loadings** we can recover actionable information from the PCA informed decision boundaries.



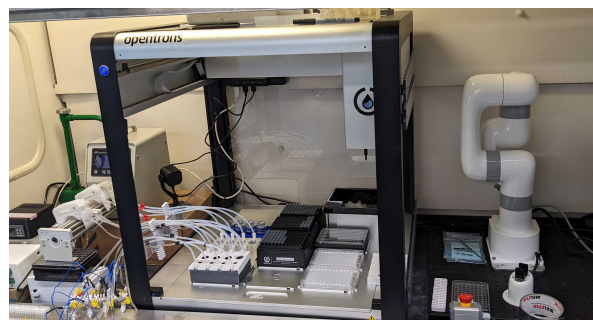
Robotic Autonomous Imaging Surface Evaluator In this project I worked with a team to build an autonomous robotic platform for surface chemistry. This involved writing a **socket server** and **client** to pass data between instruments. Image analysis for droplet contouring was completed using a **CNN**. This information was then passed to a **multi-objective bayesian optimizer** to recommend the next set of parameters for minimizing contact angle and cost.



raw droplet image



measuring contour



robotic lab station